GenAI Medical Report Generator

Automating clinical documentation using Generative AI

Course: Generative AI-2 | **Submitted To:** Mr. Jagat Pal

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Problem Statement

Healthcare professionals spend significant time manually creating clinical documentation. This time-consuming process leads to documentation delays, inconsistent report formats, and reduced patient interaction time.

Manual Documentation

Doctors spend 30-45 minutes per patient creating detailed reports

Format Inconsistency

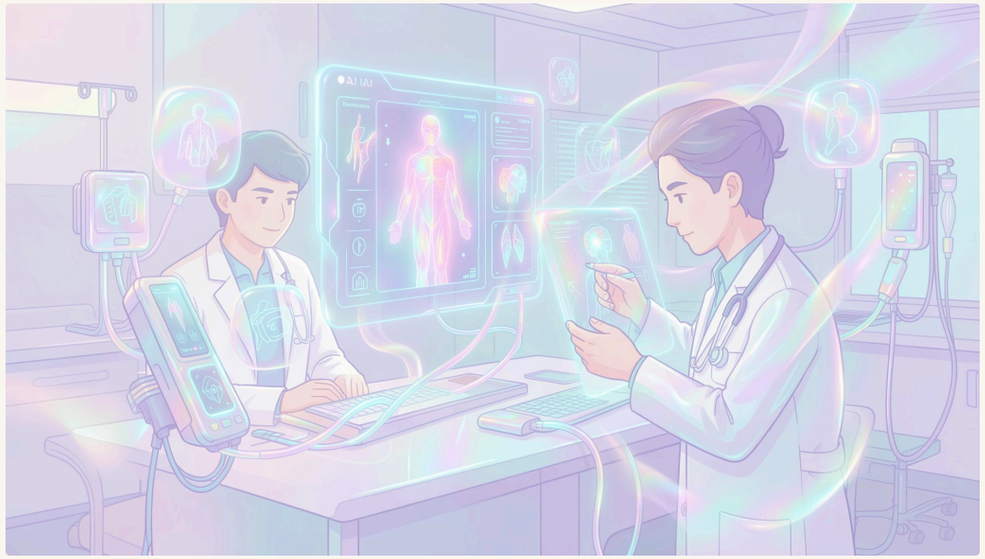
Variations in report structure across different practitioners

Efficiency Gap

Need for rapid, standardized, AI-assisted report generation

Project Objective

Develop an intelligent system that transforms patient clinical data into structured medical reports using advanced Generative AI. The solution aims to streamline documentation workflows while maintaining clinical accuracy and standardization.



01

Automate Report Generation

Convert patient data into structured clinical documentation instantly

02

Ensure Consistency

Maintain standardized SOAP format across all generated reports

03

Improve Efficiency

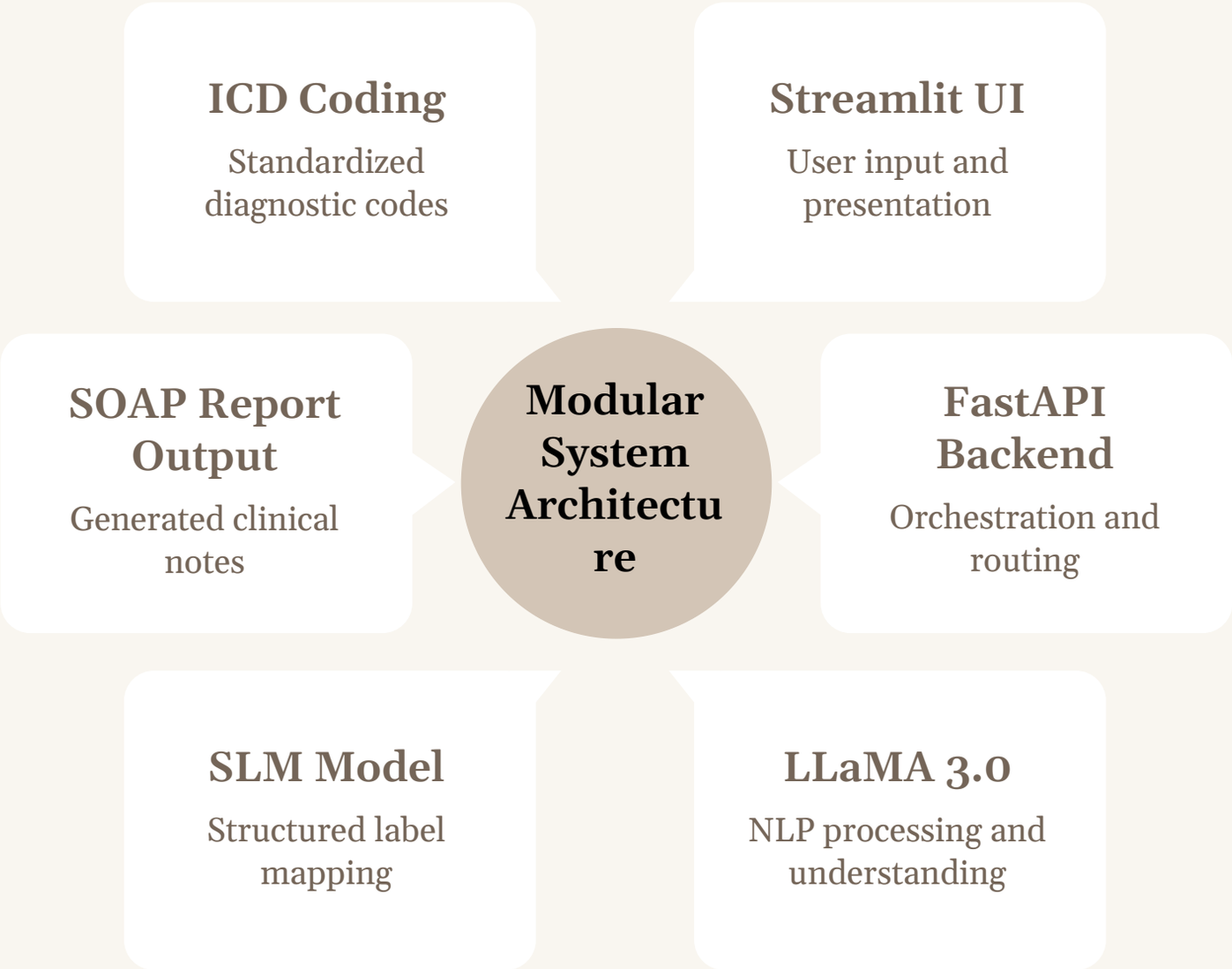
Reduce documentation time from 45 minutes to under 2 minutes

04

Enhance Usability

Provide intuitive interface accessible to all healthcare professionals

System Architecture



Technologies Used

Python

Core programming language for backend development and AI model integration

FastAPI

High-performance backend framework for building RESTful APIs

Streamlit

Framework for creating interactive web-based user interface

LLaMA 3.0

Advanced large language model for comprehensive report generation

Small Language Model

Lightweight AI for fast, low-resource deployment scenarios

VS Code

Integrated development environment for coding and debugging

Key Features

Patient Data Input

Comprehensive form capturing demographics, vitals, complaints, and medical history

SOAP Generation

Automatically structures reports in Subjective, Objective, Assessment, Plan format

ICD Codes

Intelligent suggestions for International Classification of Diseases codes

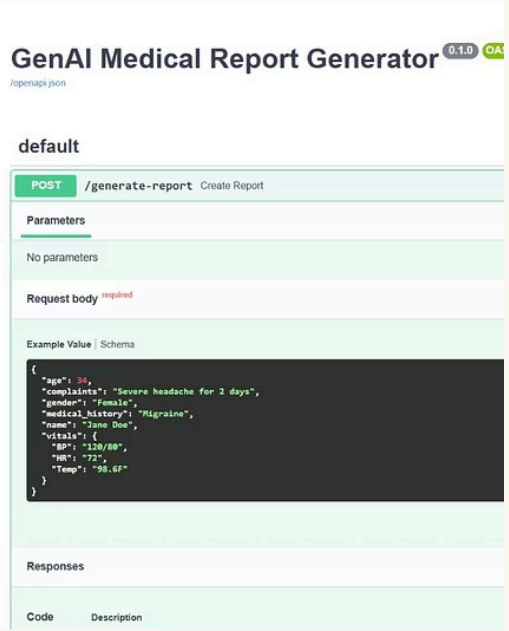
Fast Processing

Backend processes data and generates reports within seconds

Project Screenshots

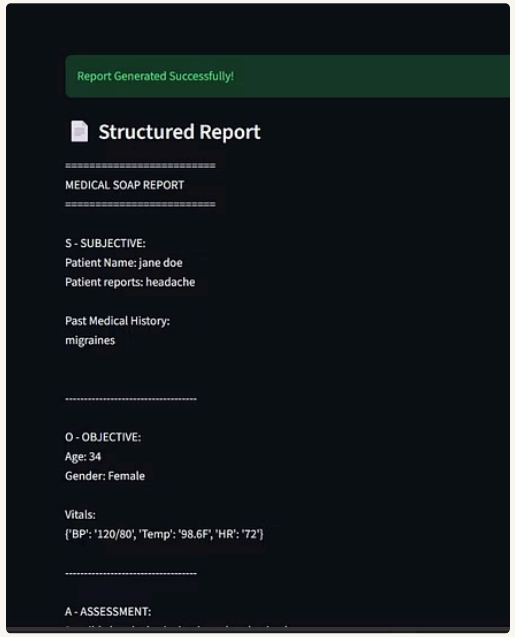
GUI Interface

[Project screenshot: GUI Interface]



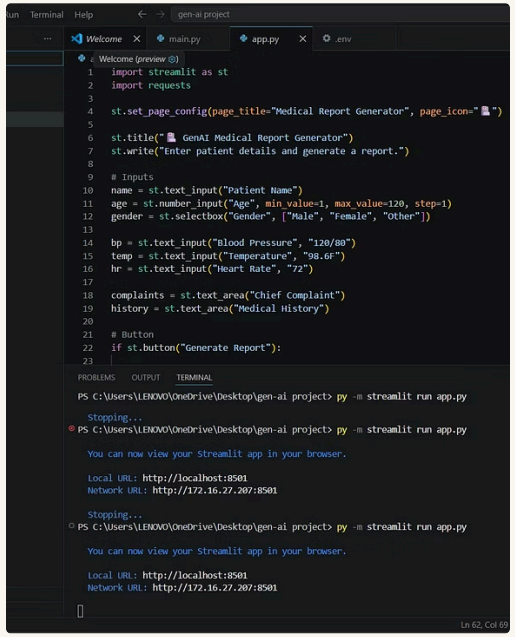
Backend API

[Project screenshot: Backend API]



Generated Output

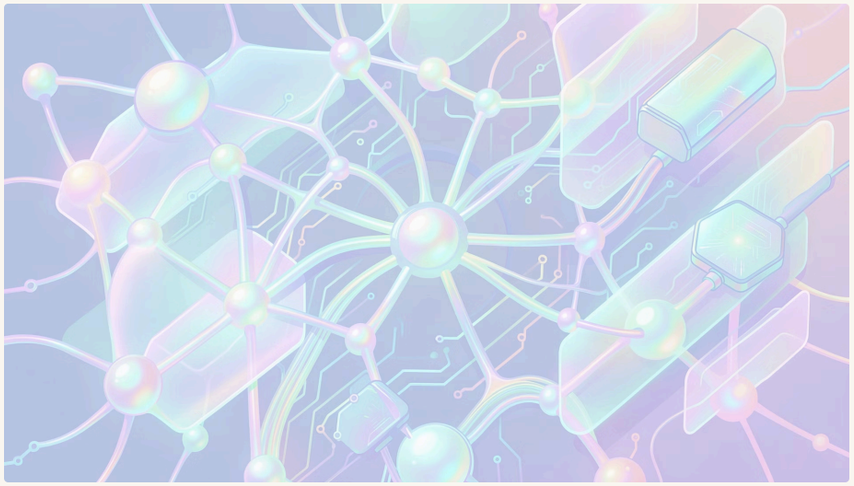
[Project screenshot: Generated SOAP Report]



Code

[Project screenshot: VS Code code]

Benefits of SLM + LLaMA 3.0



Combining lightweight SLM deployment with powerful LLaMA 3.0 architecture delivers optimal performance across different environments while maintaining clinical accuracy.

Accurate Reports

Structured clinical documentation with proper medical terminology

Faster Workflow

Reduces documentation time dramatically for healthcare providers

Scalable Architecture

Handles increased patient load without performance degradation

Lightweight Support

SLM enables offline use on low-resource systems

Future Scope

1

Hospital Integration

Connect with existing hospital management systems and EHR platforms

2

PDF Export

Direct generation of printable PDF reports with hospital branding

3

Voice System

Voice-to-report conversion for hands-free documentation during patient exams

4

Cloud Deployment

Scalable cloud infrastructure for enterprise healthcare organizations

5

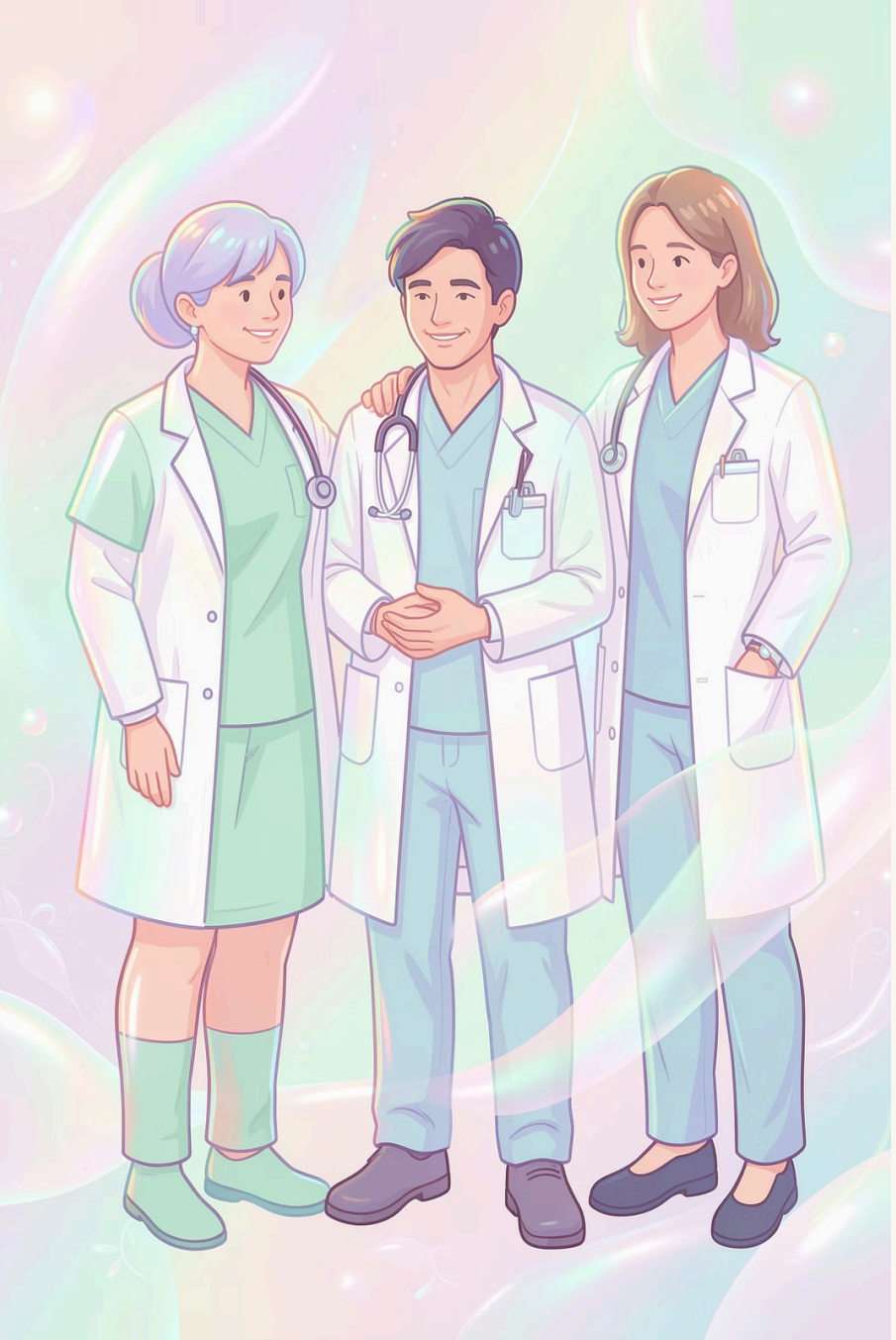
Secure Database

Encrypted patient data storage with HIPAA/GDPR compliance

6

Multilingual Support

Medical reports in multiple languages for diverse patient populations



Thank You

Questions & Answers

We're ready to discuss our GenAI Medical Report Generator implementation, technical decisions, and potential applications in healthcare settings.

Team Contact

Project team members available for detailed technical discussions

Code Repository

Full source code and documentation available for review

Demo Session

Live demonstration of system capabilities and features